# Kaafu Connect: the model and the data

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## The model in brief

Kaafu Connect is a **timetable-based journey model**: a route planner, like Google Maps transit directions, built for island boats. It isn't a machine-learning or prediction model. Nothing is trained or guessed. For a given island and a given leave-home time, it works out the earliest you can reach a hospital door using the boats that really run that day, plus the walk, bus or taxi at the end.

It has three parts:

1. **A boat network.** Every scheduled ferry leg on the chosen day (Wednesday) is a "connection" with a departure and arrival time. The routing method is a **connection scan**: go through all departures in time order and keep the earliest time each island or jetty can be reached. It's the standard method used in public-transport journey planners.
2. **A last-mile table.** For every Greater Malé jetty, hospital and hour of the day, it holds the minutes from stepping off the boat to the hospital door, by walking, bus or taxi.
3. **Observed corrections.** FollowMe GPS tracks show how long boats actually take, and how many speedboats run that aren't in any timetable. A switch on the dashboard swaps these in.

The model runs for each of 23 islands × 96 departure times (every 15 minutes) × 2 targets (nearest Greater Malé hospital, or any hospital including atoll hospitals). That's 4,416 journeys per scenario, which takes seconds. So the sliders can rerun it live.

## How one journey is calculated

Every journey is the sum of five legs: **home to harbour, wait, boat(s), getting off, and the last mile.** Here's how each is worked out.

1. **Home to harbour:** a flat 5 minutes (`ACCESS_MIN`). These islands are under about 1 km across.
2. **Wait for the boat:** the real gap between reaching the harbour and the next departure in the Wednesday timetable. If the last boat has gone, the wait runs into Thursday's timetable, which is why some trips take more than a day.

3. **Boat time:** the scheduled departure to arrival time. A multi-stop boat keeps one trip ID, so staying on board costs nothing. Changing boats needs at least 10 minutes (`BOAT_TRANSFER_MIN`).
4. **Getting off:** 5 minutes from the boat to the street (`TERMINAL_EXIT_MIN`).
5. **Last mile:** the fastest of walk, bus or taxi from that jetty to the hospital, at that hour.

The answer is the earliest hospital arrival minus the leave-home time.

## Worked example: Maafushi, leaving home at 07:00 on a Wednesday

| Leg | From → to | Minutes | Source |
|---|---|---|---|
| Home to harbour | 07:00 → 07:05 | 5 | config |
| Wait | 07:05 → 07:25 | 20 | ferry_timetable.csv (route 309 leaves 07:25) |
| Boat | 07:25 → 09:05 | 100 | route 309, Maafushi → Gulhi → Malé |
| Get off at Jetty NO5 | 09:05 → 09:10 | 5 | config |
| Walk to ADK Hospital (475 m) | 09:10 → 09:16 | 6 | facilities.csv + terminals |
| **Total** | | **136** | Over 2 hours |

With the observed switch on, FollowMe shows boats from Maafushi reaching Malé in a median 29 minutes, about 12 times a day. Most of those are speedboats. The same trip then comes in under an hour. That gap between the public ferry and what people actually use is one of the main findings.

## The routing method in one paragraph

All boat legs for Wednesday (plus Thursday, shifted by 24 hours) are sorted by departure time. Starting from the origin island at leave-home time + 5 min, the model scans each leg once. If you could be at its departure stop in time (or you're already on that boat), it records the arrival time at the next stop if that beats what it had. After one pass, it knows the earliest arrival at every island and jetty. Adding each jetty's last-mile time and taking the minimum gives the hospital arrival.

# Every dataset and how it's used

The model uses 12 sources. The boat timetable and the island and hospital lists are essential, because nothing runs without them. The rest make the last mile realistic, check the boat times, or give the benchmark.

| Dataset | What it is | Feeds | How it's used |
|---|---|---|---|
| **ferry_timetable.csv** (from MTCC/RTL CFN schedule, Aug 2026, routes 301–309, plus the Villimalé ferry schedule) | 794 boat legs with departure and arrival times and days of operation | The boat network | Filtered to Wednesday (with Thursday added for overnight waits), then turned into connections for the route planner. Gives the wait and boat legs. |
| **islands.csv** (Census 2022 via MBS Yearbook / Team Kandu, OSM harbour points) | 23 origin islands + Greater Malé, with population and harbour points | Who is counted | Population weights for every "share of people within X minutes" figure. Kaafu 17,749, inbound 13,665, total 31,414. |
| **island_names.csv** | 172 spellings mapped to one island key | Joins | Makes sure every file refers to the same island (e.g. Vaavu's Thinadhoo, not Gaafu Dhaalu's). |

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|---|---|---|---|
| **facilities.csv** (OpenStreetMap) | 9 hospitals and 20 health centres, with locations | Destinations | H01–H06 are the Greater Malé targets, and H07–H09 (Rasdhoo, Mahibadhoo, Felidhoo) are added for "any hospital". The health centres feed the "within 30 min of any health facility" indicator. |
| **terminals** (from RTL bus stops) | 4 jetties: Jetty NO5, Maafannu, Hulhumalé, Villimalé | Where boats land | Each Greater Malé boat arrival lands at one of these, and the last mile starts there. |
| **bus_service_hours.csv + bus_headways.csv** (MTCC/RTL route PDFs) | First and last bus and frequency by time of day, 15 routes | Last mile: bus | Is the bus running at that hour? Wait = half the gap between buses. |
| **RTL bus stops and departures** (Team Kandu) | Stop locations and order per route, evening trip times | Last mile: bus | Finds which route connects a jetty to a hospital, and the ride time between the two stops. |
| **Avas rides** (371,788 taxi requests, Aug–Sep 2026) | Pickup and drop-off, distance, duration, served or expired | Last mile: taxi | Taxi speed by hour, and the chance a request goes unserved, which adds retry time. |

| Dataset | What it is | Feeds | How it's used |
|---|---|---|---|
| **FollowMe GPS** (Aug 2025, 9.8 M pings in our area) | Boat positions every few minutes | Observed boat times | Pings become stops at islands, and stops become trips. This gives the real median trip time per island pair and how many boats a day, including speedboats not in any timetable. It also validates the schedule. |
| **hdx_benchmark.csv** (HDX accessibility indicators) | Car-only share of people within 30–120 min of a hospital | The comparison | The headline: "HDX says 77.4% of Kaafu within 2 h; with real boats it's Y%." |
| **MTCC.xlsx** | Ferry and bus passenger counts | Not used for findings | The numbers are randomly generated (its README says so). |
| **MACL flights** | Airport traffic | Not used | Air travel is out of scope. |

**How they connect:** islands and names give the people and the keys. The ferry timetable moves them to a jetty. The bus, Avas and terminal data move them from the jetty to a facility. FollowMe corrects the boat leg. HDX is what we compare the result against.

## Outputs and how the dashboard uses them

The model writes three tables, and the dashboard only reads these.

| Output | One row per | Dashboard use |
|---|---|---|
| `access_results.csv` | island × departure time × target | Map colours, the headline Y%, the through-the-day chart, and the trip breakdown bar (wait / boat / transfer / last mile) |
| `island_summary.csv` | island | Island panel: daytime median, best and worst, boats per day, last practical departure, scheduled vs observed gap |
| `observed_pairs.csv` | island pair × slow/fast | Scheduled vs observed comparison, and speedboat frequency for the observed switch |

**Indicators it produces:**

- Share of people within 30, 60, 90, 120 and 240 minutes of a hospital, for Kaafu and for all 23 islands, set next to HDX's car-only figures.
- The **last practical departure** per island: the latest you can leave and still reach a Greater Malé hospital the same day.
- Two MBS-style indicators with no current baseline:
  - the share of people on islands with at least one daily scheduled boat to Malé;
  - the share within 2 hours of a hospital.

**What the sliders change:** each slider edits one input and reruns the model.

- Ferry frequency and an added evening run change the boat network.
- Bus frequency and taxi availability change the last mile.
- "Thulusdhoo hospital opens" adds a destination.
- The observed switch replaces boat times with FollowMe's.

## Assumptions and limits

Every one of these goes on the limits slide. Specialist questions 1, 2, 6 and 12 test the biggest ones.

| Assumption | Value | Why it matters |
|---|---|---|
| Service day | Wednesday | Ferries alternate by day. On Wednesday, Alif Dhaalu has no Malé-bound boat. |
| Malé jetty | Jetty NO5 for all CFN ferries | Changes the walk to each hospital by up to about 15 min. |
| Boats included | Public CFN ferries + Villimalé ferry; speedboats only via FollowMe | Without the observed switch, near-Malé islands look much worse than reality. |
| Out of scope | Flights, chartered speedboats, ambulance boats | Emergencies are faster than this model; it measures routine access. |
| Walking | Straight lines at 80 m/min | Real walks are a bit longer. |
| Fixed times | 5 min home to harbour, 5 min off the boat, 10 min to change boats | Small, stated, and adjustable in config. |
| Taxi | Avas only covers rides of 2 km or more | Short taxi rides inside Malé use an average speed instead. |
| Buses | R3–R6 times come from Ramadan 2024 tables adjusted to live RTL data | Marked as estimated. |
| FollowMe | Aug 2025 data; clock is UTC (shifted +5 h); speed column in knots; not every boat has a tracker | Observed times are a sample of boat movements, not passengers. |
| HDX | Car-only model with its own population estimate | Compare percentages, not head counts. |

**In one line for the judges:** a transparent, rule-based journey planner over real timetables, checked against real boat tracks. Every number can be traced back to a timetable entry or a GPS trip.